

OPERATING SYSTEMS & SYSTEMS PROGRAMMING

COURSE PROJECT: INSTALLATION PROCESS OF PUREOS ON VMWARE WORKSTATION VIRTUAL MACHINE



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VIRTUAL OS INSTALLATION (PURE OS)

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INTRODUCTION

OPERATING SYSTEM:

- An Operating System (OS) is a collection of software that manages a computer's hardware resources and provides essential services for application programs. When you begin using a computer system, the operating system serves as an interface between the user and the computer hardware.
- The operating system is low-level software classified as system software. It supports the computer's core functions.
- An Operating System is the fundamental software that acts as an interface between the user and the computer hardware. It manages the system's resources and provides a platform on which application programs can run.
- The operating system performs essential tasks such as file management, memory management, process scheduling, handling input and output operations, and controlling peripheral devices like disk drives and printers. Without an operating system, a computer cannot function effectively or run application software.

In simple terms:

- ★ It is the main and the most important software that brings the computer to life.

PURE OS:

- PureOS is a modern Linux distribution developed by Purism and is based on the Debian operating system. It is designed with a strong focus on privacy, security, and user freedom. Unlike many other operating systems, PureOS includes only free and open-source software, ensuring transparency and reducing reliance on proprietary components.
- PureOS uses the GNOME desktop environment, providing a clean and user-friendly interface. It supports common functionalities such as

multitasking, file management, networking, and software development. Additionally, PureOS emphasizes ethical computing by protecting user data and minimizing tracking.

- In this project, PureOS is installed in a virtual environment using virtualization tools. This allows safe experimentation and understanding of operating system installation, configuration, and features without affecting the host system.

MOTIVATION:

- The main motivation behind this project is the need to learn and experiment with various concepts of the operating system in a protected manner. Rather than implementing this project on the actual machine, this is going to be done in a virtual machine in order to have an opportunity to play around with operations, configurations, and processes without causing any harm to the host system.
- Using a virtual machine in this way gives us the opportunity to work with some actual operating systems behaviors, including creating processes, inter-process communications, for instance, by using pipe() operation, executing system commands, and other important things. Using a virtual machine in this way makes it possible to configure, reset, and change things easily, which makes it possible to get some experience in this field.
- Overall, this project is motivated by the goal of bridging theoretical knowledge with practical implementation of operating system concepts using a virtual machine, while improving proficiency in Linux-based system programming.

OBJECTIVE

The main objective of this project is to gain practical understanding of operating system concepts by implementing and experimenting within a Linux environment using a virtual machine.

Specifically, the project aims to:

- Install and configure a Linux-based operating system (PureOS) in a virtual machine environment.
- Develop familiarity with system setup and basic administration in a virtualized OS environment.
- Understand and apply fundamental operating system concepts such as process creation and management.
- Explore inter-process communication (IPC) mechanisms, particularly the use of pipe() for data exchange between processes.
- Strengthen practical skills in using Linux command-line tools and system utilities.
- Bridge the gap between theoretical knowledge of operating systems and real-world system-level implementation.

Overall, the objective is to build hands-on experience with operating system functionality in a controlled virtual environment while improving system programming and Linux usage skills.

REQUIREMENTS

In order to successfully install and run PureOS in a virtual machine, the host computer should meet the following minimum Hardware and Software specifications:

Hardware Requirements:

1. **Processor (CPU):**Dual-core processor (Intel or AMD) with virtualization support (Intel VT-x or AMD-V enabled in BIOS).
2. **RAM (Memory):**Minimum 4 GB (8 GB recommended for smoother performance).
3. **Storage:**At least 20 GB of free disk space and SSD preferred for better speed.
4. **Graphics:**Basic GPU support (no high-end graphics required, but supports virtualization display acceleration)
5. **Stable power supply&Internet connection:** Having a stable internet will make the installation process efficient and seamless and we need stable power supply in order to avoid power outages.

Software Requirements:

The following software components are required to set up PureOS in a virtual environment:

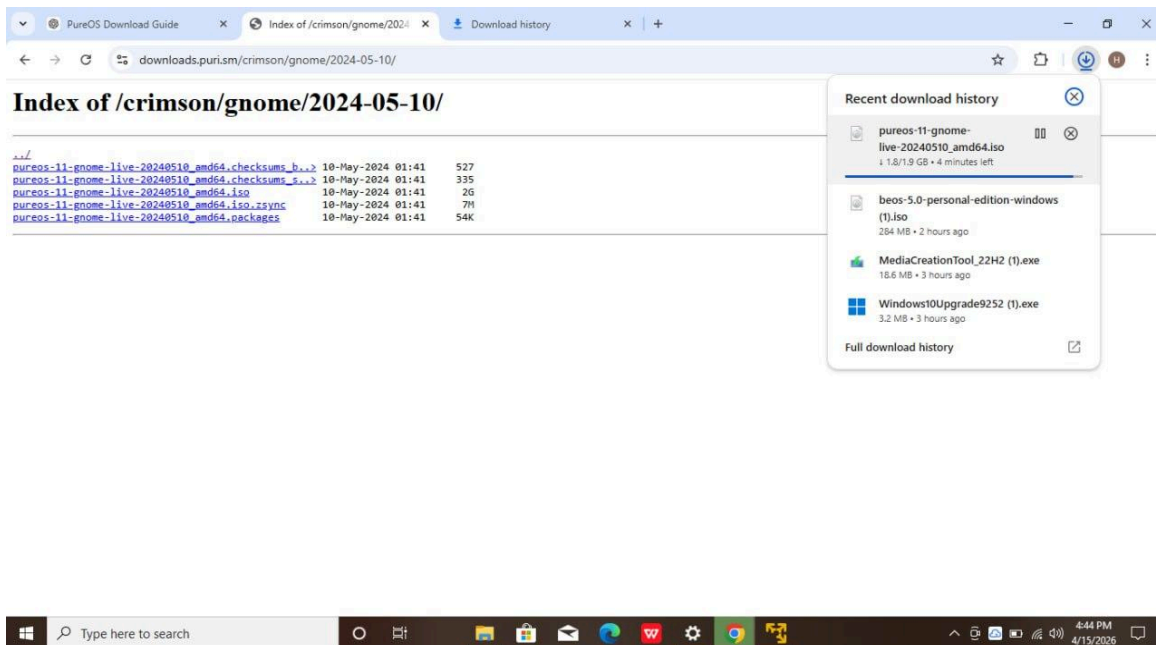
1. **Virtualization Software:**Oracle VM VirtualBox or VMware Workstation (VirtualBox is recommended for ease of use and free availability)
 2. **Operating System ISO File:**PureOS ISO image (downloaded from the official PureOS website)
 3. **TheHostOperatingSystem:**Windows,Linux,ormacOS(supports virtualization).
 4. **Additional Tools (Optional but Recommended):**VirtualBox Extension Pack for enhanced features like ,USB support Guest Additions for better display resolution and performance inside the VM.
- ★ This setup ensures that PureOS runs efficiently inside the virtual machine while allowing us to perform system programming tasks and OS-level experiments without affecting the host system.

INSTALLATION

Installation of PureOS Using VMware Workstation

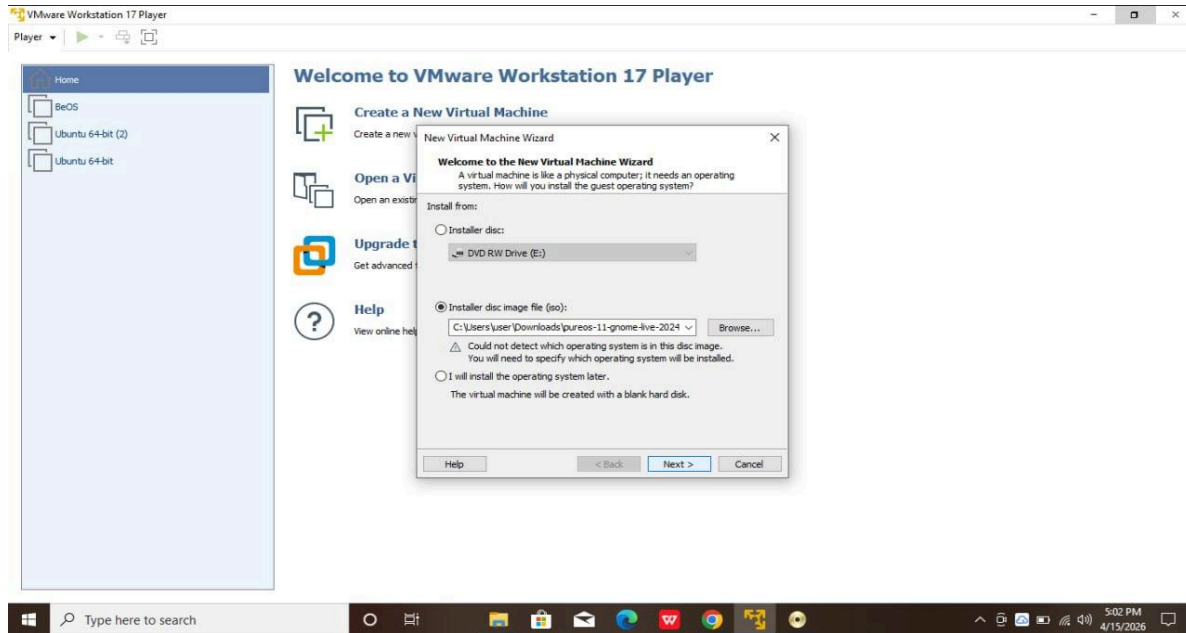
Step 1: Preparation of Required Software:

- The installation process started by preparing the necessary software tools. VMware Workstation was downloaded from the official VMware website and installed on the host computer by following the standard installation procedure. In addition, the PureOS ISO image file was downloaded from the official PureOS website. This ISO file contains all the required installation packages for the operating system and will be used as the bootable media during the virtual machine setup.



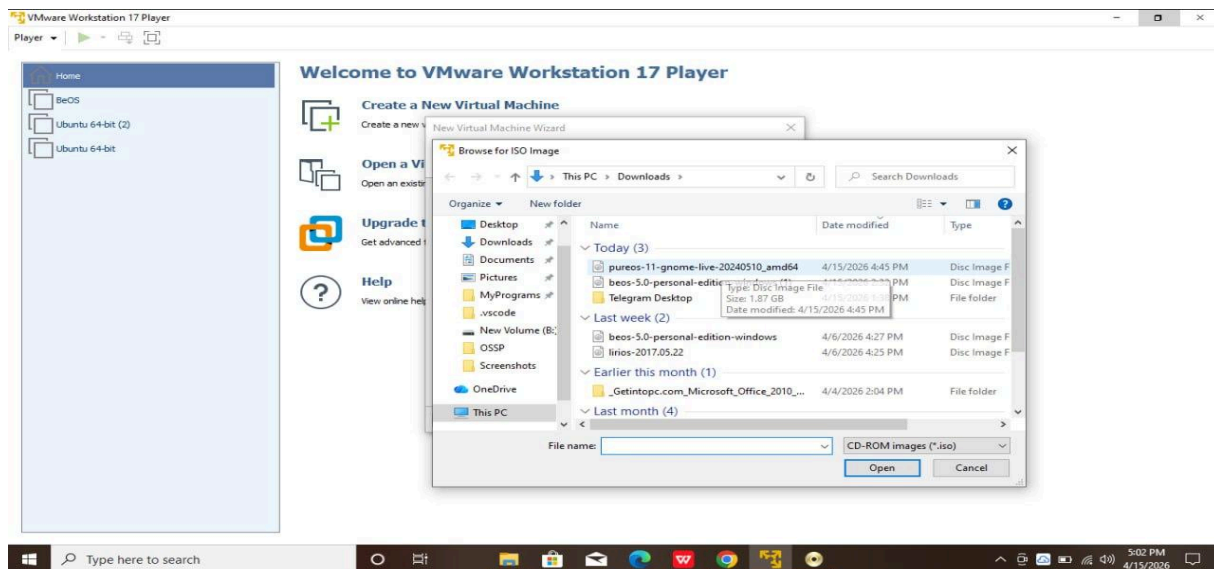
Step 2: Creation of a New Virtual Machine:

- After installing VMware Workstation, the application was launched. From the main interface, the option “Create a New Virtual Machine” was selected to begin setting up a new virtual environment. The Typical (Recommended) configuration option was chosen, as it simplifies the setup process by automatically selecting standard configuration settings suitable for most operating systems.



Step 3: Selection of Installation Media:

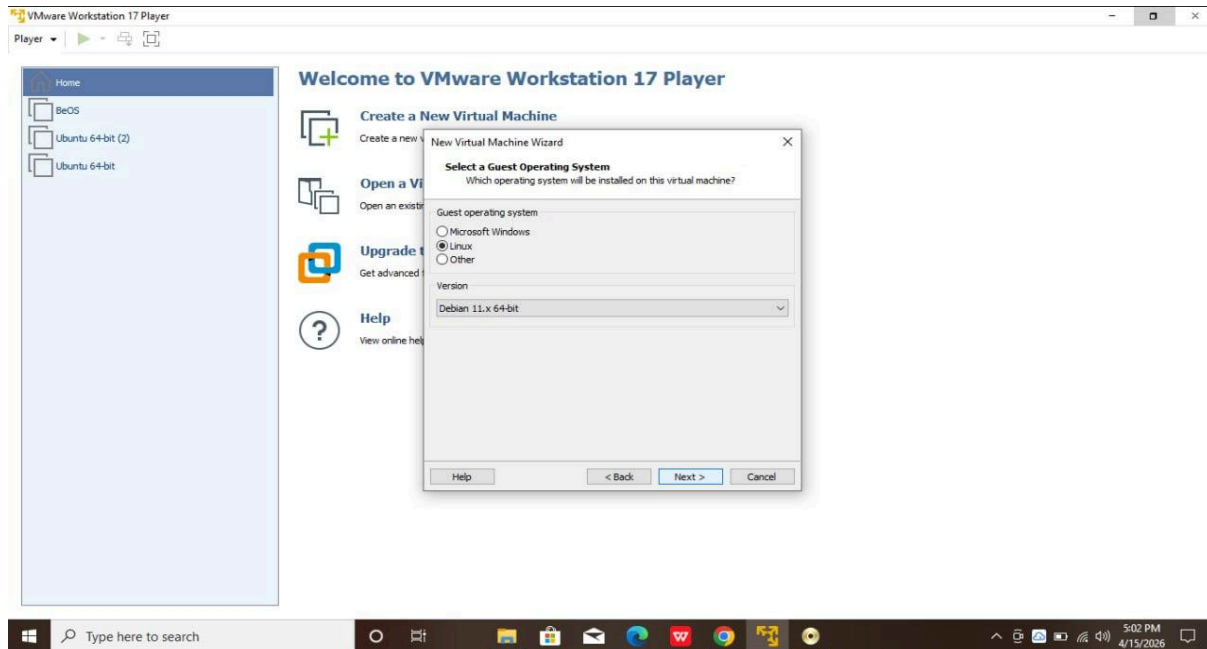
→ In this step, the installation source was specified. The option “Installer disc image file (ISO)” was selected, and the previously downloaded PureOS ISO file was located and attached. This configuration allows the virtual machine to boot directly from the ISO file and begin the operating system installation process.



Step 4: Selection of Guest Operating System:

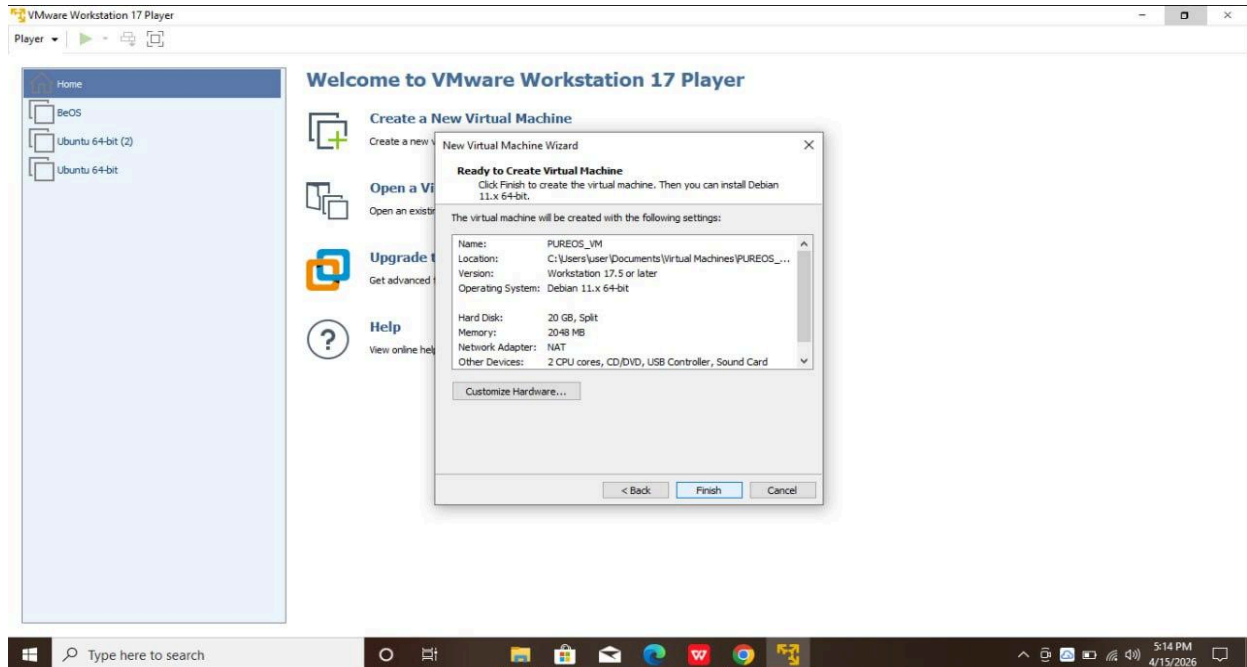
→ To ensure compatibility and optimal performance, the appropriate guest operating system type was selected. Since PureOS is based on the Debian distribution, the following options were chosen:

- **Guest Operating System: Linux**
- **Version: Debian 10/11 (64-bit):**



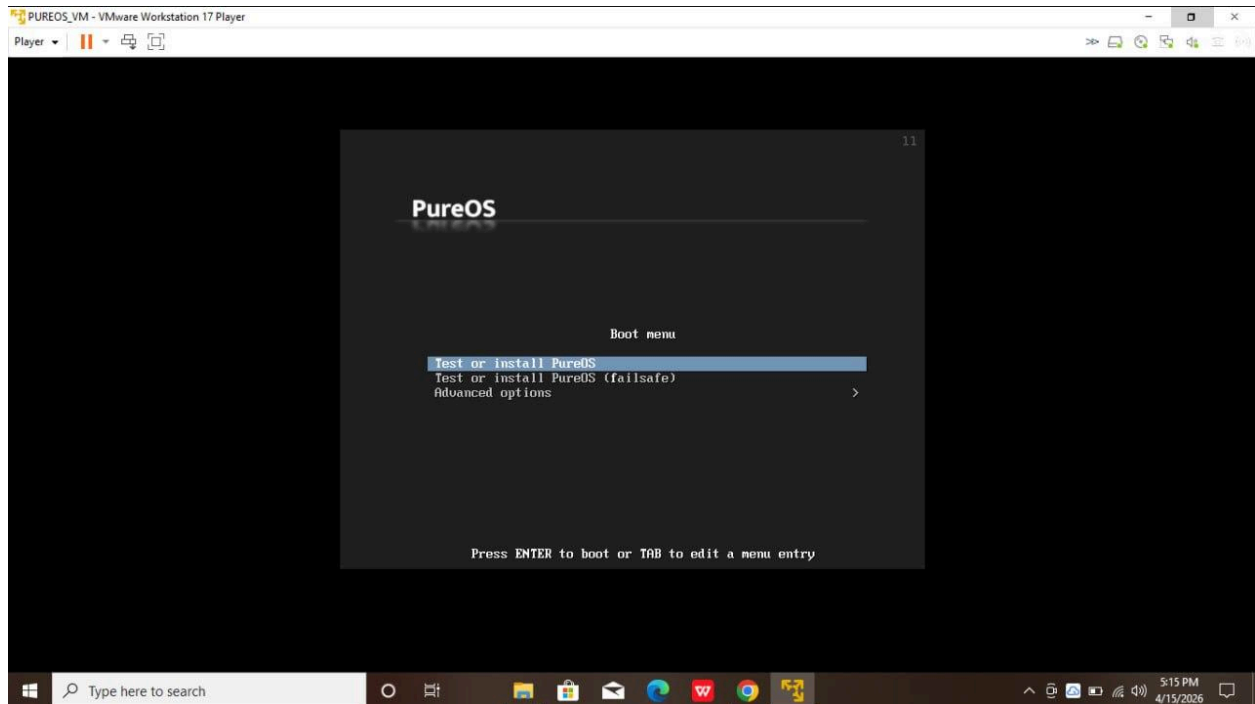
Step 5: Virtual Machine Configuration:

→ The virtual machine was then configured by assigning it a name, such as “PureOS,” and selecting a storage location on the host system. A virtual disk with a capacity of at least 20 GB was created to accommodate the operating system and additional files. Furthermore, hardware resources were customized by allocating at least 2 GB of RAM and two processor cores. These settings help ensure that the virtual machine operates efficiently.



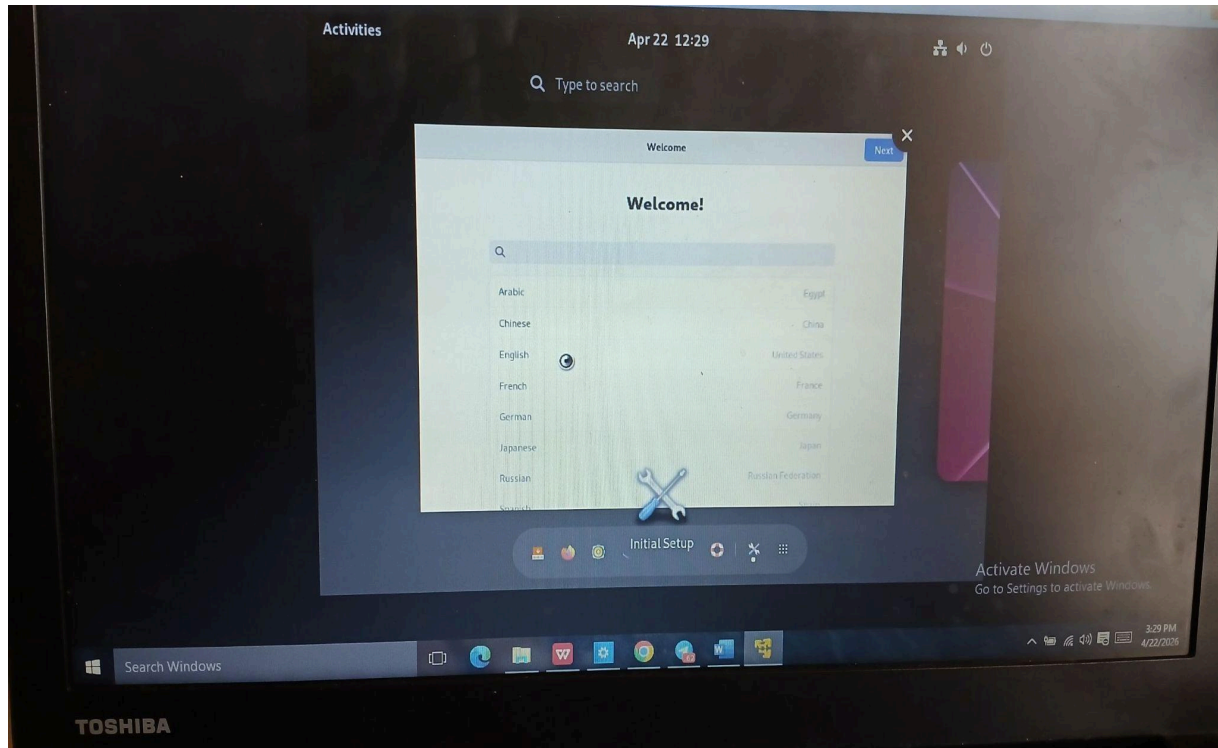
Step 6: Starting the Virtual Machine:

→ After completing the configuration, the virtual machine setup was finalized. The newly created PureOS virtual machine was selected and powered on. Upon startup, the system booted from the attached ISO file, and the PureOS installation interface was displayed.



Step 7: Operating System Installation Setup:

- The installation process was initiated by selecting the option to install PureOS from the boot menu. The installer prompted for initial configurations such as language selection and keyboard layout. Appropriate options were chosen based on user preference. The default installation settings were used to simplify the process and ensure a standard system setup.

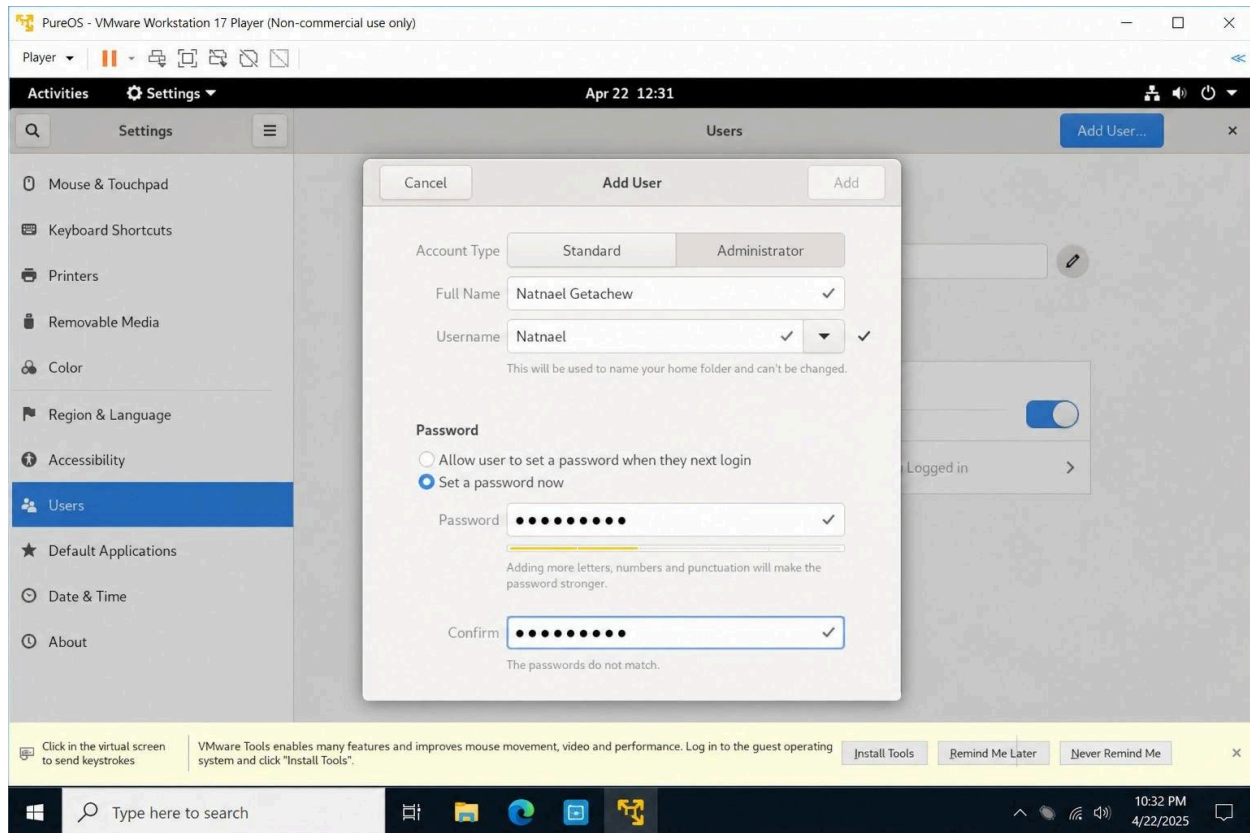


Step 8: Disk Partitioning:

- During the installation, disk partitioning was configured. The option “Erase disk and install PureOS” was selected. Since the installation was performed within a virtual machine, this option safely uses the allocated virtual disk without affecting the host system. The installer then prepared the disk for the operating system installation.

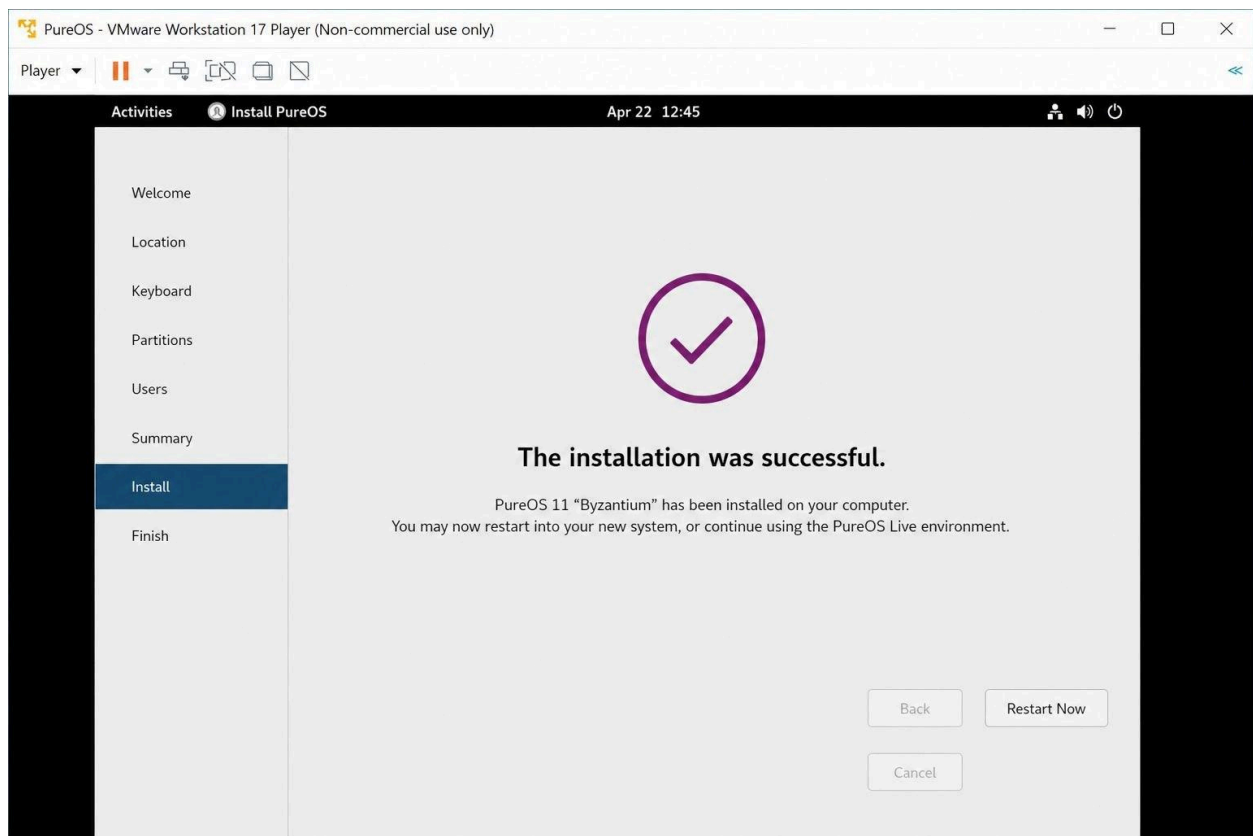
Step 9: User Account Creation:

- The next step involved creating a user account for accessing the system. The required information, including full name, username, and password, was entered. As specified in the project instructions, the full name was used appropriately. This account serves as the primary user account for system login and administration purposes.



Step 10: Installation Completion and System Login:

- After configuring the user account, the system proceeded with the installation by copying files and installing necessary components. Upon completion, the system prompted for a restart. After restarting the virtual machine, the login screen appeared.
- The user logged into the system using the previously created credentials, and the PureOS desktop environment was successfully displayed, indicating that the installation was completed successfully.



- ★ Finally the installation of PureOS using VMware Workstation was successfully completed through a structured process involving preparation, virtual machine configuration, operating system installation, and user setup. The system is now fully functional and ready for further exploration and practical use.

ISSUES AND PROBLEMS

- **Insufficient System Resources:** During the installation of PureOS on a virtual machine, inadequate allocation of system resources such as RAM, CPU cores, and storage can significantly impact system performance. PureOS, particularly when running the GNOME desktop environment, requires moderate hardware resources. When I was trying the installation, I allocated 2Gb memory resources and the installation process became slow and unresponsive.
- **ISO File Integrity Issues:** The success of the installation process depends heavily on the integrity of the PureOS ISO image. A corrupted or incomplete ISO file can lead to errors during booting or installation. These issues may include installation failures, missing system files, or unexpected interruptions during the setup process. When I first tried to use the iso file that I got from an unofficial site it failed to integrate and halted the process.
- **Network Configuration Problems:** Network connectivity is essential during installation, especially when downloading updates or additional packages. In a virtual machine environment, improper network adapter configuration may prevent internet access. This can hinder the installation process or limit system functionality after installation.
- **Compatibility Limitations of PureOS in Virtual Environments:** PureOS is not as widely optimized for virtual machine environments compared to other Linux distributions. As a result, users may experience compatibility issues such as reduced performance, limited driver support, or occasional system instability during or after installation. During my first trial of installation the system was accidentally halted and this was because of the compatibility issue.

SOLUTIONS

To mitigate the challenges discussed above, the following solutions are recommended (done):

- Allocated sufficient system resources by assigning at least 4 GB of RAM (4 GB recommended) and a minimum of 20 GB of storage.
- Ensure the ISO file is downloaded from an official source and verify its integrity using checksum validation to avoid corruption-related issues.
- Configure the virtual machine network settings appropriately, preferably using NAT mode for reliable internet access.
- Used the updated virtualization software and installed guest addition tools to enhance compatibility, improve performance, and ensure a smoother user experience.

FILE SYSTEM SUPPORT

Over View:

- ★ PureOS is a privacy-focused Debian-based Linux distribution, offers robust file support by leveraging the mainline Linux kernel. It supports popular Linux filesystems (ext4, Btrfs) for system integrity and cross-platform formats (NTFS, FAT32, exFAT) for compatibility, while supporting Mac formats (HFS+/APFS) via read-only mechanisms or third-party tools, prioritizing security and open-source compliance.

Filesystem Support in PureOS:

- ★ PureOS is a Linux distribution based on Debian, which supports a variety of filesystems to ensure compatibility, performance, and flexibility across different storage devices and operating systems. These filesystems can be categorized into native Linux filesystems and cross-platform filesystems.

1. ext4 (Fourth Extended Filesystem):

- ext4 is the default and primary filesystem used by PureOS. It is chosen because it provides a balanced combination of performance, reliability, and stability. It supports journaling, which helps prevent data corruption in case of unexpected system shutdowns. Additionally, ext4 efficiently handles large files and large storage volumes, making it suitable for modern computing environments. Its widespread use in Linux systems ensures strong community support and continuous optimization, making it the most practical and dependable choice for PureOS.

2. FAT32 (File Allocation Table 32):

- FAT32 is supported mainly for compatibility with external storage devices such as USB drives. FAT32 is widely compatible with multiple operating systems, including Windows, macOS, and Linux. This makes it ideal for transferring files between different systems. However, it has limitations such as a maximum file size of 4GB and lack of advanced security features. Despite these limitations, its universal compatibility makes it useful in PureOS environments.

3. exFAT (Extended File Allocation Table):

- exFAT is a modern replacement for FAT32 and is also supported in PureOS. exFAT removes the file size limitations of FAT32 and is optimized for flash storage devices such as USB drives and SD cards. It provides better performance and supports larger files, making it suitable for modern data transfer needs. PureOS supports exFAT to ensure compatibility with newer external storage systems.

4. NTFS (New Technology File System):

- NTFS is the default filesystem used by Windows systems and is supported in PureOS. NTFS support allows PureOS to read and write data on Windows-formatted drives. This is essential in environments where dual-boot systems or shared storage devices are used. Supporting NTFS improves interoperability between Linux and Windows systems, enabling seamless data exchange.

5. Btrfs (B-tree Filesystem):

- Btrfs is an advanced Linux filesystem that is also supported in PureOS. Btrfs provides advanced features such as snapshotting, data integrity checking, and built-in backup capabilities. These features make it suitable for systems that require high reliability and data protection. Although not the default, it is supported for users who need advanced filesystem functionalities.

6. ZFS (Zettabyte File System):

- ZFS is another advanced filesystem supported in Linux environments, including PureOS (with additional setup). ZFS is designed for high storage capacity and data integrity. It includes features such as error correction, data compression, and snapshot support. It is particularly useful in enterprise or large-scale storage systems. However, due to licensing and complexity, it is not used as a default filesystem in PureOS.

7. HFS+ (Hierarchical File System Plus):

- HFS+ is a filesystem used by older macOS systems. Support for HFS+ allows PureOS to access data from macOS-formatted drives. This is useful in environments where data is shared between Linux and older Apple systems. However, it is less commonly used today as newer macOS versions use APFS.

8. APFS (Apple File System):

- APFS is the modern filesystem used by macOS. APFS support in Linux systems, including PureOS, is limited and typically requires additional tools. This is because APFS is proprietary and optimized for Apple hardware. While PureOS may allow read access through third-party tools, full support is not natively integrated.

Overall:

- ★ PureOS supports multiple filesystems to ensure compatibility, flexibility, and efficient storage management. The default filesystem, ext4, is selected due to its reliability and performance, while other filesystems such as FAT32, exFAT, and NTFS are supported to enable interoperability with different operating systems. Advanced filesystems like Btrfs and ZFS are available for specialized use cases requiring enhanced data integrity and management features.

ADVANTAGES AND DIS-ADVANTAGES

Advantages and Disadvantages of Using PureOS on a Virtual Machine

Advantages

Safe and Isolated Testing Environment:

- Using a virtual machine allows PureOS to be installed in a completely isolated environment, separate from the host operating system. This ensures that any changes, errors, or system failures occurring within the virtual machine do not affect the host system. It is particularly useful for testing, learning, and experimentation without risking important data or system stability.

Simplified Installation and Removal Process:

- The installation of PureOS on a virtual machine does not require disk partitioning or modification of the host system's storage. This significantly reduces the complexity and risk typically associated with operating system installation. Additionally, the virtual machine can be easily removed, reinstalled, or reset, making it convenient for repeated testing or academic purposes.

Flexibility and Experimentation Capabilities:

- Virtual machines provide a high level of flexibility, allowing users to experiment with different system configurations, software packages, and updates. Features such as snapshots enable users to save the current state of the system and restore it later if needed. This makes it easier to test changes without permanent consequences.

Support for Multiple Operating Systems:

- A virtual machine enables the simultaneous use of multiple operating systems on a single physical machine. PureOS can run alongside other systems such as Windows or Linux distributions, allowing users to compare features, test compatibility, and perform cross-platform tasks efficiently.

Controlled Resource Allocation:

- Virtualization software allows users to allocate specific amounts of CPU, memory, and storage to the virtual machine. This controlled allocation helps in managing system resources effectively and ensures that the host system remains operational while running the virtual environment.

Disadvantages

Reduced System Performance:

- PureOS running on a virtual machine generally experiences lower performance compared to installation on physical hardware. This is because system resources are shared between the host and the virtual machine. Performance limitations are more noticeable in tasks that require high processing power or advanced graphical capabilities.

High Resource Requirements:

- Running a virtual machine demands a significant amount of system resources, including RAM, CPU, and disk space. On systems with limited hardware capacity, this can lead to slow performance not only within the virtual machine but also on the host system.

Limited Access to Physical Hardware:

- Virtual machines do not always provide full access to the host system's hardware components. Certain devices, such as advanced graphics cards, USB peripherals, or specialized hardware, may have limited functionality or may not be fully supported within the virtual environment.

Compatibility and Optimization Challenges:

- PureOS is not as widely optimized for virtual environments as some other operating systems. As a result, users may encounter issues such as graphical glitches, driver incompatibilities, or occasional system instability during operation.

Dependence on Host Operating System:

- The operation of the virtual machine is entirely dependent on the host system. Any failure, performance issue, or shutdown of the host operating system will directly impact the availability and functionality of the PureOS virtual machine.

CONCLUSION

- The installation and use of PureOS on a virtual machine provides a practical and efficient approach for testing, learning, and experimentation without affecting the host operating system. Throughout the installation process, several challenges may arise, including insufficient resource allocation, ISO file integrity issues, network configuration problems, and compatibility limitations within virtual environments. However, these challenges can be effectively managed through proper configuration and adherence to recommended setup practices.
- Despite certain performance and hardware limitations, virtualization offers significant advantages such as system isolation, flexibility, and ease of management. It enables users to explore the features and functionality of PureOS in a controlled environment, making it particularly suitable for academic purposes, system analysis, and initial deployment testing. Overall, using PureOS on a virtual machine is a reliable and safe method, provided that the system requirements and configurations are carefully considered.

FUTURE OUTLOOK AND RECOMMENDATION

- ★ To enhance the overall experience of installing and using PureOS on a virtual machine, several improvements and best practices can be considered. First, allocating sufficient system resources, including memory, processing power, and storage, is essential to ensure stable performance. As virtualization technology continues to evolve, future systems with more advanced hardware support will further reduce performance gaps between virtualized and native environments.
- ★ It is also recommended to use updated virtualization software such as the latest versions of VirtualBox or VMware, as these provide improved compatibility, better driver support, and enhanced graphical performance. The installation of guest addition tools should be considered a necessary step to optimize display, input devices, and system responsiveness.
- ★ From a system perspective, future developments in PureOS may focus on improving compatibility with virtual environments by enhancing driver support and optimizing system performance. Users are encouraged to keep their system updated and follow official documentation to benefit from ongoing improvements.
- ★ In addition, for more advanced use cases, users may explore alternative virtualization solutions or lightweight desktop environments to achieve better performance. Continuous learning and experimentation with virtual machine configurations will also help users gain deeper technical understanding and improve system efficiency.

Virtualization in Modern Operating Systems (What, Why, and How)

What is Virtualization:

- ★ Virtualization in modern operating systems is a technology that enables a single physical machine to run multiple independent computing environments simultaneously. These environments are called virtual machines (VMs). Each virtual machine acts like a complete computer system with its own operating system, applications, storage, and network settings, even though they are all running on the same physical hardware.
- ★ This is made possible through a software layer called a hypervisor, which creates and manages virtual machines. The hypervisor abstracts the physical hardware and presents virtual hardware resources to each virtual machine, making them operate as if they are separate physical systems.

Why Virtualization is Used:

- ★ Virtualization is widely adopted in modern computing because it improves efficiency, flexibility, and cost-effectiveness. Instead of requiring multiple physical computers, organizations can run several operating systems on a single machine, reducing hardware costs and space requirements.
- ★ It is also very important for testing and development environments, where users can safely install and test software without affecting the main system. If something goes wrong, the virtual machine can simply be reset or deleted without harming the host system.
- ★ Another key reason is system isolation and security. Each virtual machine operates independently, so problems such as system crashes, malware, or configuration errors in one VM do not directly impact others or the host operating system. Virtualization also supports backup and recovery mechanisms, such as snapshots, which allow users to save system states and restore them later if needed.

How Virtualization Works:

- ★ Virtualization works through a specialized software component called a hypervisor, which manages communication between the physical hardware and virtual machines. The hypervisor is responsible for allocating system resources such as CPU, RAM, storage, and network access to each virtual machine in a controlled and efficient manner.
- ★ When a virtual machine is created, the hypervisor simulates a complete hardware environment for it. The guest operating system installed inside the VM believes it is running on real physical hardware, even though it is actually using virtual resources provided by the hypervisor.

There are two main types of hypervisors:

→ **Type 1 (Bare-metal hypervisor):**

This type runs directly on the physical hardware without requiring a host operating system. It is commonly used in enterprise environments and data centers because it provides high performance and efficiency.

→ **Type 2 (Hosted hypervisor):**

This type runs on top of an existing operating system such as Windows or Linux. It is commonly used on personal computers for learning, testing, and development purposes. Examples include VirtualBox and VMware Workstation.

Overall:

- ❖ Through this mechanism, virtualization allows multiple operating systems to run simultaneously, share hardware resources, and function independently while being managed by a single physical machine.